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Course code : CSA1613

Course Name : Data warehousing and
Data Mining.

Assessment Tool-2

CO-1

* Topic-1: Decision support System:

* A DSS is a computer system that helps people make better decisions.

Need :

- * Helps in decision making.
- * Saves time
- * Reduces mistakes
- * Improves work.

Components:

- * Data: Information.
- * Model: Analyzes data
- * User Interface: used by the user.

Types :

- * Data-driven
- * Model-driven
- * Knowledge-driven

Applications:

- * Banking: Loan approval
- * Healthcare: Disease diagnosis

Advantages:

- * Fast decisions
- * Better Planning
- * Easy data analysis.

ETL:

- * Extract
- * Transform
- * Load.

Data Mart:

A small Part

Benefits:

- * Better dec
- * Faster ref
- * Stores old

TOPIC-4 :

OLAP

helps analys

data cube

stores data

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* Topic-2: Operational system vs DSS

OLTP

used for daily analysis

OLAP

used for data analysis

Differences:

- * Operational System: Daily work
- * DSS: Decision Making.

Performance:

- * Operational: Fast transactions
- * DSS: Fast analysis.

uses:

- * Operational: ATM, billing
- * DSS: Sales reports

Integration:

Operational systems provide data to DSS.

* Topic-3: Data warehouse:

- * A Data warehouse stores data from different sources in one place.

Three layers:

- * Data source
- * Data warehouse
- * Reports.

ETL:

- * Extract
- * Transform
- * Load.

Data Mart:

A small Part of a data warehouse

Benefits:

- * Better decisions
- * Faster reports
- * Stores old data.

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TOPIC-4 : OLAP and Data Cubes :

OLAP

Helps analyze large amounts of Data

Data cube

Stores data in many dimensions.

Operations:

- * Slice
- * Dice
- * Roll-up
- * Drill-down

Applications:

- * Sales analysis
- * Business reports
- * Finance.

* Topic-5: Dimensional Data Modeling

Fact table

Stores numbers like sales.

Dimension Table

Stores details like Product and Customer.

Star schema

Simple and Fast

Snowflake schema

More organized but Complex

Example

Sales database with Product, Customer, and time.

* Topic-6: Query Processing & BI Tools

Query Processing

* Read query

* Optimize

* Execute

* Show result

Query Optimization

* Makes queries faster

BI Tools

* Tools for analyzing data.

Examples:

- * Power BI
- * Tableau
- * KNIME

Reporting

Shows data using charts and graphs.

Open-source BI Tools:

- * KNIME
- * Metabase
- * Apache Superset.

* Topic-7: OLAP using KNIME

KNIME

An open-source data analysis tool.

Work flow

uses nodes and connections.

OLAP operations

- * slice
- * Dice
- * Roll-up
- * Drill-down

Example:

Analyze sales data

Report Generation:

Creates reports and charts.